

Samuel Farrens

Research Director, CEA Paris-Saclay

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Nationality: Irish **Languages:** English (Native), Spanish (Fluent), French (B2/C1), Italian (Basic)

Work Experience

- Jan 2026 – present
Commissariat à l'Énergie Atomique (CEA) Paris-Saclay, France
Research Director
Projects: Euclid, AstroDeep, TITAN, TOSCA, UNIONS | Management: Euclid OU-LE3 co-lead; 3x2pt Pipeline Group co-lead
- Dec 2017 – Dec 2025
Commissariat à l'Énergie Atomique (CEA) Paris-Saclay, France
Staff Researcher
Projects: Euclid, ARGOS, AstroDeep, COSMIC, DEDALE, TITAN, TOSCA, UNIONS | Management: Euclid OU-LE3 WL co-lead, COMB-CL lead, 3x2pt Pipeline Group co-lead; ARGOS WP7 lead; COSMIC/UNIONS software maintainer
- Oct 2015 – Nov 2017
Commissariat à l'Énergie Atomique (CEA) Paris-Saclay, France
Postdoctoral Researcher
Funding: DEDALE FET H2020 | Projects: DEDALE, Euclid | Lead developer: PySAP, ModOpt
- Nov 2013 – Sep 2015
Osservatorio Astronomico di Trieste (INAF-OATS), Italy
Postdoctoral Researcher
Funding: ASI – Euclid | Projects: Euclid
- Dec 2010 – Oct 2013
Institut de Ciències de L'Espai (ICE-IEEC), Spain
Postdoctoral Researcher
Funding: CosmoComp FP7 ITN Fellowship (Marie Curie Actions) | Projects: DES, Euclid, PAU

Education

- Dec 2024
Université Paris-Saclay (UPS), France
Habilitation à Diriger des Recherches (HDR)
Thesis: “Advanced tools and software development for cosmology”
- Oct 2007 – Nov 2010
University College London (UCL), UK
PhD in Astrophysics
Thesis: “Optical Detection of Galaxy Clusters”
- Sep 2003 – Jun 2007
University College London (UCL), UK
MSci in Astrophysics — First Class
Thesis: “Finding Clusters with 10^6 Luminous Red Galaxies”
- Sep 2002 – Jun 2003
Birkbeck College, University of London, UK
Foundation Degree in Mathematics and Physics — Distinction
- Jan 1999 – Dec 2001
Liceo de Atenas, Costa Rica
International Baccalaureate Diploma — Distinction

Select Presentations

- Dec 2025
[Invited] University of Turin Seminar, Turin, Italy
“Probing the Dark Universe with Euclid”
- Sep 2025
[Invited] ASTRODAT, Durham, UK
“Scientific Software Development”

May 2025	[Invited] BDL3, Paris, France “Using AI to Probe the Invisible Universe”
Apr 2023	[Invited] ADA X, Hersonissos, Greece “Scientific Software Development”
Apr 2023	[Invited] ICDSS, Hefei, China “ShapePipe: Open source weak lensing for modern surveys”
Jul 2022	EuroPython, Dublin, Ireland “ShapePipe: A modular weak-lensing pipeline”
Sep 2021	[Invited] Artificial Intelligence 2021 (remote) “Python Sparse data Analysis Package”
May 2021	[Invited] Where the Earth Meets the Sky (remote) “Hands-On Image Processing with PySAP”
Nov 2020	[Invited] ADASS XXX, Granada, Spain (remote) “PySAP: From Galaxies to Brains and Beyond”
Sep 2019	[Keynote] EuroScipy, Bilbao, Spain “From Galaxies to Brains! – Image processing with Python”
May 2017	[Invited] Université Clermont Auvergne Seminar, Clermont-Ferrand, France “Regularisation techniques for solving inverse problems”

Supervision

PhD Students

Jan 2026 – present	Sammy Sharief Co-supervised with François Lanusse and Tobias Liaudat
Oct 2023 – present	Ezequiel Centofanti Projects: ARGOS, Euclid Thesis: “ <i>Machine learning methods for radio and optical imaging in astronomy</i> ” Co-supervised with Jean-Luc Starck

Postdoctoral Researchers

Oct 2024 – present	Emma Ayçoberry Projects: ARGOS, Euclid
Oct 2024 – present	Cail Daley Projects: Euclid, SPT, TOSCA, UNIONS Co-supervised with Martin Kilbinger
Oct 2024 – present	Calum Murray Projects: Euclid, UNIONS Co-supervised with Martin Kilbinger
Feb 2022 – May 2024	Lucie Baumont Projects: Euclid, Rubin LSST, UNIONS Co-supervised with François Lanusse and Martin Kilbinger
Nov 2020 – Aug 2022	André Zamorano Vitorelli Projects: Euclid, UNIONS Co-supervised with François Lanusse and Martin Kilbinger

Research Engineers

Mar 2025 – present	Chaitanya Chawak Project: Euclid Topic: Euclid OU-LE3 data model management and data-driven PSF model
Nov 2023 – present	Nada Moukaddem Project: Euclid Topic: Euclid data-driven PSF model and cluster mass estimation
Nov 2022 – present	Jennifer Pollack Project: Euclid Topic: PSF model integration into Euclid Science Ground Segment

Apr 2021 –
Apr 2024 **Sergio Miranda La Hera**
Project: Euclid | Topic: Euclid OU-LE3 data model management and software

Master's Students

Apr–Sep
2026 **Maxime Ronceray** “Automatically Differentiable Image Simulations for Euclid” | Co-supervised with François Lanusse

Apr–Sep
2026 **Vincent Brunet** “Galaxy shape inference with machine learning methods” | Co-supervised with Emma Ayçoberry

Mar–Aug
2025 **Manal Bensahli** “Machine Learning for Radio Interferometric Array Design” | Co-supervised with Ezequiel Centofanti

Mar–Aug
2023 **Alex Szapiro** “Star Selection for PSF Modelling with Deep Learning”

Apr–Aug
2021 **Olivier Roth** “Unsupervised learning for galaxy cluster membership assignment” | Co-supervised with Joana Frontera Pons

Sep 2019 –
Feb 2020 **Sophie Starck** “PySAP Software Development and Data Science”

Jul–Dec 2019 **Alice Lacan** “Deep Learning for Multiclass Classification of Blended Galaxy Images”

Jul–Dec 2019 **Joël Gehin** “Euclid Data Processing”

Apr–Sep
2018 **Alexandre Bruckert** “Machine Learning and Image Deblending for Astrophysical Data”

Nov 2017 –
Apr 2018 **Hugo De Bouard** “Déconvolution d’images de galaxies”

Jul–Dec 2017 **Xinyu Wang** “Identification of Blended Galaxy Images” | Co-supervised with Jean-Luc Starck

Apr–Sep
2017 **Peipei Ran** “Deconvolution of astrophysical images with space variant instrumental response” | Co-supervised with Jean-Luc Starck

Teaching

Jan 2023 –
present UE PRAT M2 Master's Course, Sorbonne Université, Paris, France
Lecture: “A practical introduction to signal processing in astrophysics” (3.5 h/year)

Aug 2023 –
present École d’été Rodolphe Clédassou, France
Lecture: “Scientific Software Development for Euclid” (3 h/year)

Jul 2021 EuroPython (remote)
Tutorial: “The Anatomy of a Python Class” (1.5 h)

Nov 2019 Euclid C++/Python Workshop, Paris, France
Tutorials on Python and Make/CMake (10 h)

Sep 2019 EuroScipy, Bilbao, Spain
Tutorial on Astronomical Image Processing (1.5 h)

May 2018 ADAIX Summer School, Valencia, Spain
Tutorial on Sparsity (5 h)

May 2016 ADA8 Summer School, Chania, Greece
Tutorial on Sparsity (3 h)

Jan & May
2011 ICE, Barcelona, Spain
Lectures on cosmology for master's students (5 h)

Oct 2008 – Apr 2010	UCL, London, UK Marking: 1st Year Classical Mechanics, Thermodynamics, 3rd Year Practical Physics
Oct 2008 – Apr 2009	UCL, London, UK Tutoring undergraduate Maths and Physics

Event Organisation

Jan 2026	Euclid OU-LE3 Meeting, Bologna, Italy — Role: SOC
Jun 2025	COLOURS, Orsay, France — Role: Primary organiser, SOC and LOC
May 2025	Euclid 3x2pt Meeting, Paris, France — Role: LOC
Jun 2024	Euclid Consortium Meeting, Rome, Italy — Role: SOC
May 2024	COSMO21 Conference, Chania, Greece — Role: LOC
Jan 2024	Euclid France Meeting, Saclay, France — Role: LOC
Sep 2023	ADA X Summer School, Hersonissos, Greece — Role: LOC
Mar 2023	DAP GPU Half-Day, Saclay, France — Role: SOC and LOC
May 2018	ADA IX Summer School, Valencia, Spain — Role: Primary organiser, SOC and LOC
May 2018	COSMO21 Conference, Valencia, Spain — Role: LOC

Community Activities

Ongoing	Peer Review: Astronomy & Astrophysics (A&A), MNRAS, Computing in Science and Engineering
Apr 2024	PhD Jury — External <i>vocal</i> reviewer, Universitat Autònoma de Barcelona (Computer Science)
2018–2020	Project Review — Swiss National Science Foundation

Computing Skills

Programming	Python, C/C++, TensorFlow, Matlab/Octave, IDL, HTML, CSS/SASS
Build & DevOps	Make, CMake, pyproject.toml, GitHub Actions, GitLab CI/CD, Docker, Singularity
HPC & Databases	MPI, OpenMP, PBS, Slurm, GPU Computing, SQL
Machine Learning	Neural Networks, CNNs, SVMs
Distribution	PyPI, Conda, Homebrew
Docs & Testing	Sphinx, Doxygen, $\LaTeX$, PyTest, Catch2

Observing

Sep 2007	Australian Astronomical Observatory (AAO) Instruments: 3.9m telescope with the AAOmega spectrograph
Feb 2006	Observatoire de Haute-Provence (OHP) Instruments: 1.52m + AURELIE spectrograph; 1.2m + CCD camera

Outreach

Dec 2015	Workshop on Galaxy Zoo at Les Savanturiers, France
Mar 2010	Demonstrator at Your Universe, UCL festival of astronomy, UK